PAUL WOLUJEWICZ, PH.D., M.P.H.

Assistant Professor of Medical Sciences
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ACADEMIC APPOINTMENTS & RESEARCH AFFILIATIONS

Assistant Professor of Medical Sciences *September 2026 – Present*
Department of Medical Sciences, Frank H. Netter MD School of Medicine
Joint Appointment: Assistant Professor of Biomedical Sciences
Department of Biomedical Sciences, School of Health Sciences
Quinnipiac University, Hamden, CT

Assistant Professor of Biomedical Sciences (Tenure-Track) *July 2023 – September 2026*
Department of Biomedical Sciences, School of Health Sciences
Quinnipiac University, Hamden, CT

Adjunct Assistant Professor of Neuroscience *July 2023 – Present*
Brain and Mind Research Institute, Weill Cornell Medicine, New York, NY

CAES Research Affiliate *November 2025 – Present*
The Connecticut Agricultural Experiment Station (CAES), New Haven, CT

EDUCATION

Ph.D., Physiology, Biophysics & Systems Biology *2022*
Weill Cornell Graduate School of Medical Sciences, New York, NY
Thesis: A genomic characterization of human neural tube defect risk

M.S., Neuroscience *2015*
Rutgers Biomedical and Health Sciences, Newark, NJ

M.P.H., Biostatistics & Epidemiology *2013*
Rutgers School of Public Health, Piscataway, NJ

B.S., Biophysics *2008*
University of Scranton, Scranton, PA

HONORS & AWARDS

Outstanding Faculty Scholar Award *April 2026*
Quinnipiac University

Outstanding Mentor Award — Scholarly Reflection and Concentration Capstone (SRCC) *March 2025*
Frank H. Netter School of Medicine, Quinnipiac University

Timothy M. George Award for Excellence in Neural Tube Defect Research *November 2022*
12th International Neural Tube Defects Conference

Weill Cornell Community Affairs Distinguished Service and Leadership Award *April 2019*
Weill Cornell Medical College

NIH T32 Training Grant in Developmental and Stem Cell Biology *2018–2019*
Weill Cornell Medical College

NIH Big Data Coursework for Computational Medicine (BDC4CM) *2017*
Weill Cornell Medicine, University of Minnesota, and Johns Hopkins University

Rutgers Biomedical and Health Sciences GSBS Research Award *May 2015*
Batish Lab, Rutgers University

RESEARCH INTERESTS

Neurogenomics of neurodevelopmental and neurogenetic disorders (neural tube defects, multiple sclerosis, Krabbe disease), including genetic variant architecture, gene-environment interactions, and archaic introgression; and vector-borne disease biology, with a focus on tick-borne pathogen metagenomics.

GRANTS & FUNDING

SHS Faculty Scholarship Grant *2026–2027*
"Characterizing Anaplasma Phagocytophilum Variant in Connecticut Blacklegged Ticks" — PI

SHS Faculty Scholarship Grant *2025–2026*
"Determining Transcriptional Programs Underlying Aberrant Neural Tube Closure" — PI

Faculty Scholarship & Creative Works Impact Grant *2024–2025*
"Functional Genomics Underlying Birth Defects of the Nervous System" — Co-PI (with Carter Takacs)

SHS Faculty Scholarship Grant *2024–2025*
"Leveraging Metagenomics for Tick-Borne Pathogen Surveillance" — PI

Pending Extramural Applications

NIH/NICHD R15 *2026*
"Dissecting Digenic Inheritance in Neural Tube Defects Using Functional Genomics and Zebrafish Models"
Contact PI (Multi-PI with Carter Takacs)

PEER-REVIEWED PUBLICATIONS

** Mentored trainee co-author*

1. Stankovic, I., Smit, P.*, Cross, J., Rai, A., **Wolujewicz, P.**, Greening, D., & Colak, D. (2025). Extracellular vesicle profiling reveals novel autism signatures in patient-derived forebrain organoids. *Translational Psychiatry*, 15(1), 393. https://doi.org/10.1038/s41398-025-03607-w
2. **Wolujewicz, P.**, Aguiar-Pulido, V., Thareja, G., Suhre, K., Elemento, O., Finnell, R. H., & Ross, M. E. (2024). Integrative computational analyses implicate regulatory genomic elements contributing to spina bifida. *Genetics in Medicine Open*, 2, 101894. https://doi.org/10.1016/j.gimo.2024.101894
3. Stankovic, I., Notaras, M., **Wolujewicz, P.**, Lu, T., Lis, R., Ross, M. E., & Colak, D. (2024). Schizophrenia endothelial cells exhibit higher permeability and altered angiogenesis patterns in patient-derived organoids. *Translational Psychiatry*, 14(1), 53. https://doi.org/10.1038/s41398-024-02740-2
4. Crane-Smith, Z., De Castro, S. C. P., Nikolopoulou, E., **Wolujewicz, P.**, Smedley, D., Lei, Y., Mather, E., Santos, C., Hopkinson, M., Pitsillides, A. A., Genomics England Research Consortium, Finnell, R. H., Ross, M. E., Copp, A. J., & Greene, N. D. E. (2023). A non-coding insertional mutation of Grhl2 causes gene over-expression and multiple structural anomalies including cleft palate, spina bifida and encephalocele. *Human Molecular Genetics*, 32(17), 2681–2692. https://doi.org/10.1093/hmg/ddad094
5. Allen, M., Huang, B. S., Notaras, M. J., Lodhi, A., Barrio-Alonso, E., Lituma, P. J., **Wolujewicz, P.**, Witztum, J., Longo, F., Chen, M., Greening, D. W., Klann, E., Ross, M. E., Liston, C., & Colak, D. (2022). Astrocytes derived from ASD individuals alter behavior and destabilize neuronal activity through aberrant Ca2+ signaling. *Molecular Psychiatry*, 27(5), 2470–2484. https://doi.org/10.1038/s41380-022-01486-x

6. Aguiar-Pulido, V., **Wolujewicz, P.**, Martinez-Fundichely, A., Elhaik, E., Thareja, G., Abdel Aleem, A., Chalhoub, N., Cuykendall, T., Al-Zamer, J., Lei, Y., El-Bashir, H., Musser, J. M., Al-Kaabi, A., Shaw, G. M., Khurana, E., Suhre, K., Mason, C. E., Elemento, O., Finnell, R. H., & Ross, M. E. (2021). Systems biology analysis of human genomes points to key pathways conferring spina bifida risk. *Proceedings of the National Academy of Sciences*, 118(51), e2106844118. https://doi.org/10.1073/pnas.2106844118
7. **Wolujewicz, P.**, Steele, J. W., Kaltschmidt, J. A., Finnell, R. H., & Ross, M. E. (2021). Unraveling the complex genetics of neural tube defects: From biological models to human genomics and back. *Genesis*, 59(11), e23459. https://doi.org/10.1002/dvg.23459
8. **Wolujewicz, P.**, Aguiar-Pulido, V., AbdelAleem, A., Nair, V., Thareja, G., Suhre, K., Shaw, G. M., Finnell, R. H., Elemento, O., & Ross, M. E. (2021). Genome-wide investigation identifies a rare copy-number variant burden associated with human spina bifida. *Genetics in Medicine*, 23(7), 1211–1218. https://doi.org/10.1038/s41436-021-01126-9
9. Chapman, L. M., Spies, N., Pai, P., Lim, C. S., Carroll, A., Narzisi, G., Watson, C. M., Proukakis, C., Clarke, W. E., Nariai, N., Dawson, E., Jones, G., Blankenberg, D., Brueffer, C., Xiao, C., Kolora, S. R. R., Alexander, N., **Wolujewicz, P.**, Ahmed, A. E., Smith, G., … Zook, J. M. (2020). A crowdsourced set of curated structural variants for the human genome. *PLoS Computational Biology*, 16(6), e1007933. https://doi.org/10.1371/journal.pcbi.1007933
10. **Wolujewicz, P.**, & Ross, M. E. (2019). The search for genetic determinants of human neural tube defects. *Current Opinion in Pediatrics*, 31(6), 739–746. https://doi.org/10.1097/MOP.0000000000000817
11. Felling, R. J., Covey, M. V., **Wolujewicz, P.**, Batish, M., & Levison, S. W. (2016). Astrocyte-produced leukemia inhibitory factor expands the neural stem/progenitor pool following perinatal hypoxia-ischemia. *Journal of Neuroscience Research*, 94(12), 1531–1545. https://doi.org/10.1002/jnr.23929
12. Dhaliwal, A., Brenner, M., **Wolujewicz, P.**, Zhang, Z., Mao, Y., Batish, M., Kohn, J., & Moghe, P. V. (2016). Profiling stem cell states in three-dimensional biomaterial niches using high content image informatics. *Acta Biomaterialia*, 45, 98–109. https://doi.org/10.1016/j.actbio.2016.08.052

Manuscripts in Preparation & Under Review

13. **Wolujewicz, P.**, Assi, B.*, Khalil, N., Matijevic, J.*, Rogers, E., Eggers, C. H., Molaei, G. A Feasibility Study of Nanopore Adaptive Sampling for Multi-Pathogen Detection in *Ixodes scapularis*. *Vector-Borne and Zoonotic Diseases (under review)*
14. Narayanan, S.*, Pettengill, K.*, Ren, X., Takacs, C., Ross, M.E., Hersh, D., **Wolujewicz, P.** Mapping the Genetic Architecture of Open Spina Bifida: A Catalog of the Rare Variants Reported in Human Studies. *(in preparation)*

PRESENTATIONS & CONFERENCES

Society for Birth Defects Research and Prevention — 66th Annual Meeting | *June 2026*
"Trio-Based Whole Genome Sequencing Reveals Inherited Rare Structural Variation in the Genetic Architecture of Spina Bifida" | Washington, DC (platform)

NEURON Conference | *April 2026*
"AI & Systems Neuroscience" — Workshop Speaker | Hamden, CT

Krabbe Translational Disease Network 2026 | *March 2026*
Translational Science (invited roundtable panelist)

CDC Vector Week | *March 2026*
"Evaluating Nanopore Metagenomic Sequencing for Real-Time Surveillance of Vector-Borne Pathogens" | poster

Society for Birth Defects Research and Prevention — 65th Annual Meeting | *July 2025*
"AI in Genomics" — Invited Speaker & Mini-course Lead | Denver, CO

NEURON Conference | *April 2025*
"AI Tools in Neurobiology" — Workshop Speaker | North Haven, CT

13th International Conference on Neural Tube Defects | *August 2024*
"Evidence of Pathogenic Expansions in Myelomeningocele Case-Parent Trios" | platform

Cell Symposia: Engineering Development and Disease in Organoids | *August 2024*
"Validation of digenic interactions underlying human neural tube defects using single rosette neural tube organoids from isogenic iPSCs" | poster

Society for Neuroscience Annual Meeting | *October 2024*
"Behavioral Effects and Transcriptional Signatures of CD11b+ Microglia in Chronic Unpredictable Stress" | poster

ACTRIMS Forum (Americas Committee for Treatment and Research in MS) | *February 2023*
"Polygenic Risk Scores Associate with Lesion Features on Quantitative MRI" | poster

12th International Conference on Neural Tube Defects | *October 2022*
"The Regulatory Landscape Underlying Human Neural Tube Defects" | platform

Center for Neurogenetics, Weill Cornell Medicine | *March 2022*
"Bioinformatics and Variant Calling for Genetic Counselors" — invited

Englander Institute for Precision Medicine, Weill Cornell Medicine | *January 2020*
"Advancing Clinical and Research Pipelines for Whole Genome Sequencing Data"

International Conference on Neural Tube Defects — Harvard Medical School | *September 2019*
"A Global View of Rare Copy Number Variation in Human Spina Bifida" | Boston, MA (platform)

Regeneron Science to Medicine, Regeneron Pharmaceuticals | *June 2019*
"From Whole Genome Sequencing Data to Networks in Neurodevelopmental Disorders" | Tarrytown, NY

American Society of Human Genetics (ASHG) Annual Meeting | *October 2018*
"Rare Copy Number Variation in Congenital Malformations of the CNS" | San Diego, CA (poster)

NTD International Summit, Weill Cornell Medicine | *February 2018*
"Standardizing High-Confidence Bioinformatic Pipelines for Variant Discovery"

NYU Nature Neurogenetics Symposium, New York University | *August 2017*
"Detecting Structural Variation from Whole Genomes: Benchmarking and Pipeline Optimization" | poster

PROFESSIONAL EXPERIENCE

Postdoctoral Associate in Neuroscience *September 2022 – June 2023*
Brain and Mind Research Institute, Weill Cornell Medical College

- Served as primary computational genomics specialist for the Center for Neurogenetics (CNG), performing variant interpretation across pediatric and adult cases
- Collaborated with neurologists, genetic counselors, and pathologists to integrate genomic findings with clinical phenotypes
- Developed and deployed AI/ML and systems biology pipelines for variant detection, pathogenicity prediction, and structural variant interpretation across neurogenetic disorders; authored comprehensive genomic interpretation reports for complex neurodevelopmental and neurodegenerative conditions

Ph.D. Research *June 2018 – August 2022*
Weill Cornell Graduate School of Medical Sciences
Mentors: Dr. Margaret Elizabeth Ross (Director, Center for Neurogenetics) & Dr. Olivier Elemento (Director, Englander Institute for Precision Medicine)

- Conducted multi-omic and computational analyses to identify genomic risk factors for neural tube defects, integrating population genomics, regulatory genomics, and systems-level network analysis

Doctoral Research Rotations *August 2016 – May 2018*
Weill Cornell Graduate School of Medical Sciences & Memorial Sloan Kettering Cancer Center

- Hagen Tilgner Lab (WCM) — developed wet- and dry-lab protocols for single-nuclei sequencing and computational long-read isoform analysis
- Alex Kentsis Lab (MSKCC) — genome-wide translocation sequencing (HTGTS) to detect DNA double-strand breaks

Research Teaching Specialist IV *May 2014 – August 2016*
New Jersey Medical School, Department of Microbiology, Biochemistry and Molecular Genetics
Mentor: Dr. Mona Batish

- Conducted advanced molecular biology research on Spinach2 RNA aptamer systems for live-cell RNA imaging applications
- Developed and optimized single-molecule fluorescence in situ hybridization (smFISH) protocols for cancer research
- Optimized gene reporter and nuclear protein assays with imaging techniques to predict mesenchymal stem cell differentiation via intracellular dynamic organization

Senior Laboratory Technician *August 2009 – May 2014*
RUCDR – Infinite Biologics, Rutgers University, Piscataway, NJ

- Managed large-scale nucleic-acid processing projects funded by NIDDK, NIDA, COGA, and NIMH
- Performed nucleic-acid extractions and automated DNA/RNA distribution via liquid-handling robotics
- Conducted high-throughput gene-expression assays on microarray platforms

TEACHING & CURRICULUM DEVELOPMENT

Curriculum Innovation — *designed and implemented new courses for medical and biomedical science programs*

- **Machine Learning in Medical Sciences** (BMS622 / MED740) — *Spring 2026*
- **Computational Biomedicine** (BMS519 / MED700) — *Spring 2025, Fall 2026*
- **Biomedical Genomics** (BMS312) — *Fall 2024, 2025, 2026*
- **Neurogenetics** (BIO500) — *Summer 2024*

Course Instruction

- **Cellular Basis of Neurobiological Disorders** (BMS578) — *Fall 2025, 2026*
- **Biotechnology** (BMS472) — *Spring 2024, 2025, 2026*
- **Seminar in Healthcare Disparities** (BMS556) — *Spring 2025, 2026*
- **Genetics** (BIO282) — *Fall 2023, 2024, 2025, 2026*
- **Immunology & Immunology Lab** (BMS522/L) — *Fall 2023, 2024*
- **Neurobiology** (BIO329) — *Spring 2023, 2024*
- **Biology Journal Club** (BIO250) — *Fall 2023*

External Guest Lectures

- **Weill Cornell Medicine — Genetic Counseling Graduate Program (2026):** invited lectures in *Molecular Methods, Bioinformatics and Variant Analysis for Clinicians* (MGC009) — "Advances in Genomic Testing" and "Raw to Processed Data for Variant Analysis"

Prior Teaching Appointments

- **Part-Time Faculty, Quinnipiac University, Dept. of Biology (2021–2023):** General Biology I (BIO101), Biology Journal Club, and Neurogenetics
- **Adjunct Lecturer, John Jay College of Criminal Justice (2019–2021):** Bioinformatics for Genomics & Biomedical Data Science (BIO380, course developer); Modern Biology (BIO101/BIO102)
- **Teaching Assistant, Weill Cornell Medicine (Fall 2018):** Critical Dissection of Scientific Data
- Biostatistics Tutor, Rutgers School of Public Health (2012–2013)

MENTORING

Student Mentoring (2023–Present)

- **Master's Thesis Students:** Michael Ragazzi (GBMS, 2026-2027); Jesse Matijevic (MCB, 2026-2027); Natalie Grober (GBMS, 2025-2026; co-mentored with Carter Takacs); Brandon Assi (GBMS, 2024-2025)
- **Medical Student Capstone Mentoring (SRCC, Netter School of Medicine):** Sanjana Narayanan (2026-present); Shaila Gopal (2026-present; co-mentored with Zaenab Dhari); Johanna Appleton (2026-present; co-mentored with Zaenab Dhari); Alan Chai (2024-present); Phil Smit (2023-2026); John Gribbin (2022-2025)
- **QUIP-RS (Quinnipiac Interdisciplinary Program for Research and Scholarship):** Jesse Matijevic (2025); Jenna Visich (2025); Kaylee Pettengill (2024)

Thesis Committee Service

- **Master's Thesis Committees, Quinnipiac University:** Giada Mercuriano; Isabelle Miserani; Jessica Cyr; Vlora Alka; Payton Prete; Weronika Lewna

Mentored Student Presentations

- Puar, M., Etts, K., Schlack, R., et al. *Identifying Patterns of Inherited Rare Variant Risk in Human Myelomeningocele Case-Parent Trios Leveraging Transmission Disequilibrium and Digenic Inheritance Testing*. Society for Birth Defects Research and Prevention (BDRP) Annual Meeting, Washington, DC — poster, June 2026
- Chodkiewicz, A. *Early-Life Stress Induces Gene Expression Changes in the Rat Hippocampus*. NEURON Conference — poster, April 2026
- Grober, N. *Defining Neurodevelopmental Roles of HNRNPU Through CRISPR/Cas-Mediated Knockdown and Transcriptomic Analysis*. NEURON Conference — poster, April 2026
- Triana, R. *A Case-Control Volumetric Analysis of Hippocampal Atrophy in Relapsing-Remitting Multiple Sclerosis*. NEURON Conference — poster (selected for data blitz), April 2026
- Assi, B. *Metagenomics for Tick-Borne Pathogen Detection*. Yale Vector-Borne & Zoonotic Diseases Symposium — lightning talk, February 2026
- Assi, B., et al. *Investigating the Feasibility of Nanopore Metagenomic Sequencing for Real-Time Vector-Borne Pathogen Surveillance*. NMCA Annual Conference — platform, December 2025
- Smit, P., et al. — *Transcriptomic Profiling from Extracellular Vesicle Autism Patient-Derived Forebrain Organoids Reveals Novel Signatures*. American Society of Human Genetics (ASHG) Annual Meeting — platform, October 2025
- Chai, A., et al. — *Exploring the Role of Short Tandem Repeat Expansions in the Genetic Etiology of Neural Tube Defects*. ASHG Annual Meeting — poster, October 2025
- Etts, K., Schlack, R., et al. — *Examining Digenic Rare Variant Combinations Contributing to Neural Tube Defects (NTDs)*. ASHG Annual Meeting — poster, October 2025
- Schlack, R., & Etts, K. — *"Unveiling Digenic Interaction in CNS Structural Birth Defects."* NEURON Conference, North Haven, CT — poster (selected for data blitz), April 2025
- Gribbin, J., Wolujewicz, P., & Soda, K. J. — *"X-Chromosome Analysis in Multiple Sclerosis for Identification of Genes Implicated in Female-Biased Disease Presentation."* American Academy of Neurology (AAN) Annual Meeting, Denver, CO — platform, April 2024

PROFESSIONAL SERVICE

- Science Committee, Society for Birth Defects Research and Prevention (BDRP) | *2026 – Present*
- Co-Editor (invited), Teratology Primer, 4th Edition, BDRP — in progress | *2026 – Present*
- Session Chair, NEURON Conference — AI workshop sessions | *2025 – 2026*
- **Ad Hoc Reviewer:**
  Genetics in Medicine, Birth Defects Research, PLOS ONE, Scientific Reports, Epigenetics, BMC Pediatrics

UNIVERSITY SERVICE

- Faculty Fellow, School of Business, User Experience and Behavioral Insights Lab | *Fall 2026 – Spring 2027*
- Co-Editor, SHS Bobcat Byte (AI-tips newsletter for School of Health Sciences faculty and staff) | *Fall 2026 – Present*
- QU Health Professions Advisory Committee (HPAC) | *Spring 2026 – Present*
- Dean's Generative AI in Healthcare Education Task Force | *Fall 2025 – Present*
- Path A Faculty Search Committee, School of Health Sciences | *2025*
- New Faculty Kickoff Panelist — "What I Wish I Knew as a New Faculty Member" | *August 2025*
- Graduate Biomedical Sciences (GBMS) Admissions Committee | *Spring 2025 – Present*
- SHS Dean's Search Committee | *Fall 2024 – Spring 2025*
- SHS Scholarship Committee | *August 2024 – Fall 2025*
- NetGene Faculty Advisor (medical student genomics interest group) | *September 2023 – Spring 2026*

COMMUNITY SERVICE & OUTREACH

Healthcare Career Exploration Academy Summer Camp, Quinnipiac University *2023 – 2026*
Delivered interactive lesson for biomedical science (BMS) component

Invited Talk - Visiting International Student Cohort, Quinnipiac University *August 2024*
"How Neuroscience and Immunology Are Changing the World" — outreach lecture for prospective students

Medical Reserve Corps (MRC), Quinnipiac Valley Health District (QVHD) *2020 – 2023*
Public-health and community-outreach volunteer; COVID-19 vaccination clinic support

Data Science Summer Camp, Quinnipiac University *2022*
Organized content and lesson plans for a two-week on-site high-school camp; led themes in big-data visualization, data science in public health, and bioinformatics

Early STEM Outreach Club, Weill Cornell *2018 – 2021*
Cofounder & Secretary - developed and taught STEM lessons (robotics, coding, bioart) for NYC children ages 6-14

PROFESSIONAL MEMBERSHIPS

- Society for Birth Defects Research and Prevention (BDRP), Member | *2025 – Present*
- Northeast Mosquito Control Association (NMCA), Member | *2025 – Present*
- Global Alliance for Genomics & Health (GA4GH) - Genomic Knowledge Standards (GKS) Work Stream, Member | *2025 – Present*